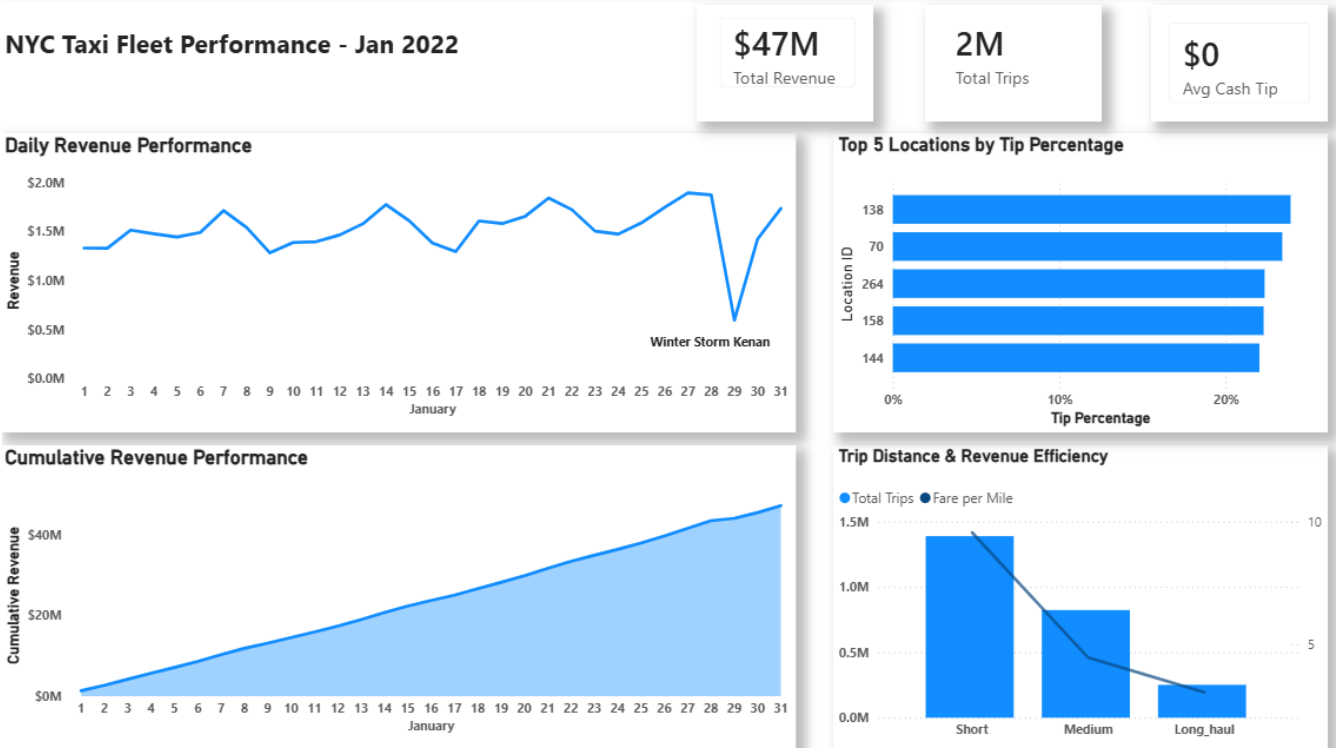
Project: NYC Yellow Taxi Trip Data



Executive Summary & Insights

- **The Challenge:**

Analyzed the January 2022 NYC Yellow Taxi dataset to evaluate daily revenue trends, trip distance efficiency, and driver tipping behavior. The goal was to use Google BigQuery and Power BI to identify operational trends supporting fleet planning and performance decisions.

- **Key Insight 1 (Weather Impact on Revenue):**

Revenue peaked at over $1.89 million on January 27, followed by a sharp 60% drop on January 29 to $596,288. This significant decline directly correlated with Winter Storm Kenan, which restricted vehicle movement.

- **Key Insight 2 (Revenue Efficiency):**

Short trips of less than 2 miles represent 56% of the volume with 1.39 million trips. These short trips generate the highest revenue efficiency at $9.55 per mile, compared to just $3.04 per mile for long-haul trips over 7 miles.

- **Key Insight 3 (Tipping & Data Quality):**

The system fails to record cash tips, systematically showing them as $0.00. For credit card payments on trips over 5 miles, LaGuardia Airport (Location 138) has the highest tip percentage at 23.8%, while JFK Airport (Location 132) yields the highest average tip amount at $9.61.

Business recommendations:

- **Recommendation 1:**

Prioritize fleet positioning for short-haul trips in urban centers to maximize per-mile revenue efficiency, while maintaining a strong presence at LaGuardia and JFK airports to capture high-tip, high-ticket fares.

- **Recommendation 2:**

Address the data quality issue regarding unrecorded cash tips to accurately track driver earnings, and implement proactive fleet management plans to minimize revenue loss during severe weather events.

- **Business Impact:**

Optimizing route focus and preparing for external weather shocks will maximize daily revenue and fleet utilization, while fixing payment data capture will provide a truer picture of driver profitability and operational health.

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Technical Execution & Data Pipeline

1.Methodology & Architecture:

To process and analyze the massive volume of NYC Yellow Taxi data, I utilized **Google BigQuery (SQL)** for data extraction, transformation, and complex aggregations. The raw dataset was queried to evaluate daily revenue trends, categorize trip distances, and calculate tipping percentages. The resulting analytical tables were then connected to **Power BI** to build an interactive, executive-level dashboard that visualizes key performance indicators (KPIs) and operational patterns.

2.SQL Analysis:

Below are the key SQL queries developed to process the dataset and extract actionable business insights:

Query 1: Daily Revenue Performance & Cumulative Total *(*Utilizing CTEs and Window Functions to track daily and running totals*)*

```
with dates as (
SELECT distinct date(pickup_datetime) AS pickup_date , sum(total_amount) as total_amount
FROM bigquery-public-data.new_york_taxi_trips.tlc_yellow_trips_2022
where pickup_datetime between '2022-01-01 00:00:00 UTC' and '2022-01-31 23:59:00 UTC'
group by  pickup_date)

SELECT dates.pickup_date , total_amount , sum(total_amount) over(order by pickup_date) as
cumulitive_total_amount
FROM dates ;
```

The complete SQL scripts, including data filtering (HAVING) and categorizations (CASE WHEN), are available on my GitHub.